LESSON PLAN

BASIC ELECTRICAL (23)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	14
Total No. of Lesson Plans	29	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Anti-skid devices, purpose and operating principle of anti-skid unit.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Anti-skid devices, purpose and operating principle of anti-skid unit.
5. Specific Objectives	23.1.3.1 Trainee will be able to describe typical alighting gears of an aircraft and its construction.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of anti-skid system diagrams; Instructor supervision required for diagram interpretation.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of the landing gear?
(b) How does a wheel rotate during landing?
(c) What happens if a wheel locks up during braking?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Purpose of anti-skid devices • Construction of typical landing gear • Role of wheel speed sensors • Hydraulic brake actuation system • Prevention of wheel lock-up	L & Ds	PPT, IB	15 min
(b)	• Operating principle of anti-skid unit • Signal processing in anti-skid logic • Modulation of brake pressure • System activation during rollout • Integration with flight control systems	L & Ds	PPT, PDF	20 min
(c)	• Failure modes of anti-skid systems • Indications of anti-skid malfunction • Maintenance checks for anti-skid units • Troubleshooting common faults • Safety implications of system failure	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how the anti-skid unit prevents wheel lock-up.
(b) Describe the sequence of events when anti-skid activates.
(c) What are the consequences of an anti-skid system failure?

12. Home Assignment and Follow Up	Prepare a brief note on the differences between typical landing gear configurations and their impact on anti-skid system design, to be discussed in LP 02.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Tracing and identification of various under carriage components, Shock observers, types, brief description and operation	Refer to the master syllabus reference list	LP 15

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min